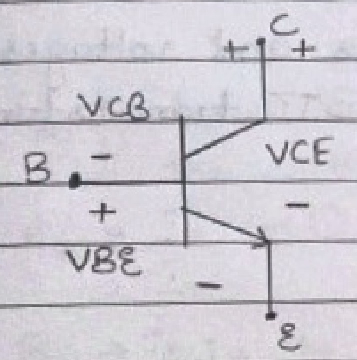
2 I A

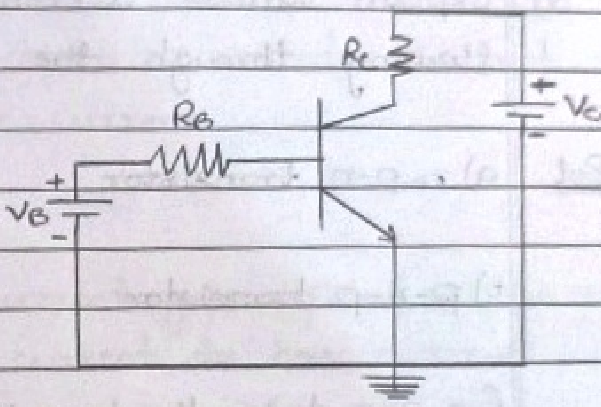
Basic Electronics

1) With a neat diagram, explain the operation of an npn transistor.

Ans The terminal voltage polarities for an npn transistor are shown in figure below



a) npn terminal voltage polarity



b) voltage source connection

-> For an npn transistor, the base is biased positive w.r.t emitter and collector is biased to higher positive voltage than base.

-> From fig (b) we know that ~~in~~ an voltage sources ~~the transistors~~ are connected to the transistors through resistors.

-> The base voltage VB is connected via resistor RB, and the collector supply Vcc is connected via ~~RB~~ Rc

-> The negative terminals of the two voltage sources are connected at the transistor emitter terminal.

-> Vcc is always much larger than VB and this ensures that the collector base junction remains reverse baised : positive on collector (n-side) and negative on base (p-side)

-> Typical transistor base - emitter voltages are similar to the diode forward voltages : 0.7V for a silicon transistor and 0.3V for a germanium device.

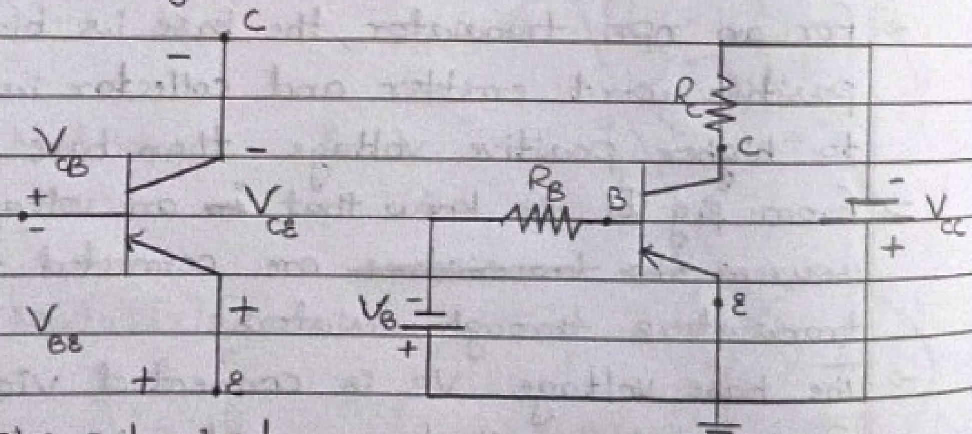
-> Collector voltages might be from 3V to 20V for many of a transistor.

2) Explain various currents and voltages flowing through the BJT transistor.

Sol a) n-p-n transistor

b) p-n-p transistor

For pnp device, the base is baised negative with respect to the emitter, and the collector is made more negative than the base.



a) pnp terminal voltage polarities   b) voltage source connection

-> Voltage sources are connected via resistors and the source positive terminals connected at the emitter

-> with V_{cc} larger than V_B, the collector is more negative than the (n-type) base and thus the collector base junction is kept reverse biased.

* All BJT's (npn & pnp) are normally operated with the collector base junction reverse biased and the base emitter junction forward biased.

3) Define α and β. Determine the relationship b/w α and β.

Sol α_{dc} → It is the ratio of collector current to the emitter current.

$$\alpha_{dc} = \frac{I_c}{I_E} \quad - (1)$$

β_{dc} → It is the current gain defined as the ratio of collector current to base current.

$$\beta_{dc} = \frac{I_c}{I_B} \quad - (2)$$

For a transistor,

$$I_E = I_c + I_B \quad - (3)$$

(as $I_c \simeq I_E$)

Dividing both sides of eq (3) by I_c we get,

$$\frac{I_E}{I_c} = \frac{I_B}{I_c} + 1$$

$$\frac{1}{\alpha_{dc}} = \frac{1}{\beta_{dc}} + 1$$

$$\alpha_{dc} = \frac{\beta_{dc}}{\beta_{dc}+1} \quad - (4)$$

(or)

$$\beta_{dc} = \frac{\alpha_{dc}}{1-\alpha_{dc}} \quad - (5)$$

Eq (4) & (5) give the relation b/w α_{dc} & β_{dc} of a transistor.

4) A transistor has $\beta = 150$ & I_E is 12mA. Calculate the approximate collector current (I_C) and base current (I_B)

Sol Given

$\beta_{dc} = 150$ $\quad I_C = ?$

$I_E = 12mA$ $\quad I_B = ?$

Since I_C is always equal to I_E

$I_C = 12mA$

$I_C = \beta \times I_B$

$I_E = I_C + I_B$

Solving for I_B

$I_C = \beta \times I_B$

$12 = 150 \times I_B$

$I_B = \frac{12mA}{150}$

$I_B = 0.08mA$

$I_C = \beta \times I_B$

$= 150 \times 0.08$

$= 12mA$

5. Calculate α_{dc} and β_{dc} for the transistor if I_C is measured as 1mA and I_B is $25\mu A$

Sol Given

$I_C = 1mA$

$I_B = 25\mu A$

$\alpha_{dc} = ?$

$\beta_{dc} = ?$

$I_C = \beta_{dc} \times I_B$

$\frac{I_C}{I_B} = \beta_{dc}$

$\frac{1 \times 10^{-3}}{25 \times 10^{-6}} = \beta_{dc}$

$\alpha_{dc} = \frac{\beta_{dc}}{\beta_{dc}+1}$

$= \frac{40}{40+1}$

$= \frac{40}{41}$

$\alpha_{dc} = 0.97$

$= 0.04 \times 10^3$

$= 40$

$\beta_{ac} = 40$

6) Describe with neat circuit diagram of BJT amplification for both voltage & current.

Sol Current Amplification

W.k.T, $I_C = \alpha_{dc} \cdot I_E$ — ①

Thus, transistor can be used for current amplification

Small change in (ΔI_B) produces large change in (ΔI_C) and a large emitter current change in (ΔI_E)

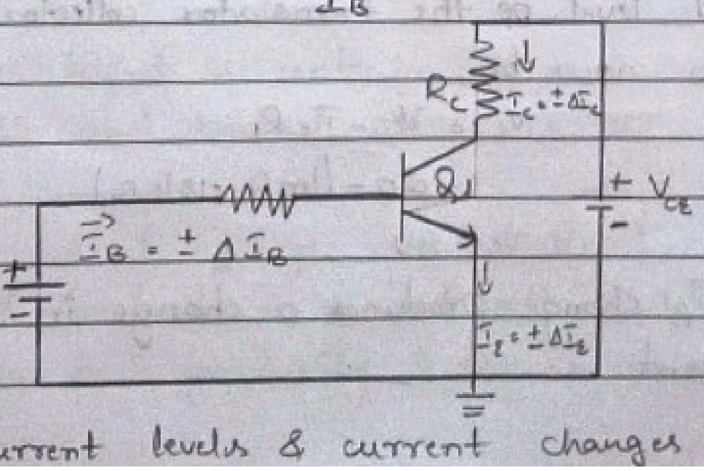
Current gain from base to collector is

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B} \quad \text{— ②}$$

Increasing & decreasing levels of input and output currents may be defined as alternating quantities.

Alternating current gain from base to collector is

$$\beta_{dc} = \frac{I_C}{I_B} \quad \text{— ③}$$

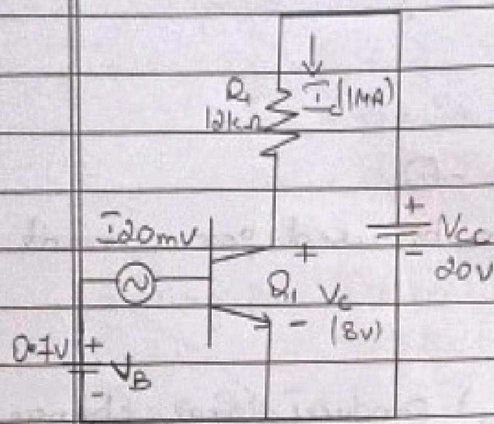


Current levels & current changes

Voltage Amplification

Assume that the transistor Q_1 has $\beta_{ac} = 50$

The 0.7Vdc voltage source (V_B) forward biases the transistor emitter base junction.



a) Transistor circuit with base bias & signal generator

b) V_{BE} changes produces I_B changes

→ AC signal source in series with V_B provides a ±20mV input voltages.

→ The transistor collector is connected to a 20V dc voltage source V_{CC} via 12kΩ collector resistor R_1

from fig (b) $I_B = 20\mu A$

$$I_C = \beta_{ac} \cdot I_B$$
$$= 50 \times 20\mu A$$
$$I_C = 1mA$$

The dc level of the transistor collector voltage can now be,

$$V_C = V_{CC} - I_C R_1$$
$$= 20 - (1mA \times 12k\Omega)$$
$$V_C = 8V$$

The I_B change produces a change in collector current

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$$\Delta I_c = \beta_{dc} \Delta I_B$$
$$= 50 \times (\pm 5 \mu A)$$
$$\Delta I_c = \pm 250 \mu A$$

fig (a) shows that ΔI_c causes a change in voltage drop across R_1 and thus produces a variation in transistor collector voltage

$$\Delta V_c = \Delta I_c R_1$$
$$= \pm 250 \mu A \times 12 k\Omega$$
$$\Delta V_c = \pm 3V$$

The voltage gain Δv is the ratio of output voltage to input voltage.

$$\Delta c = \frac{\Delta V_c}{\Delta V_B} = \frac{+3V}{\pm 20 mV}$$

$$\Delta v = 150$$

The circuit ac input is the base voltage change ΔV_B and the ac output is the collector voltage change ~~ΔA~~ (ΔV_c). Because the output is greater than the input, the circuit has a voltage gain. It is a voltage amplifier

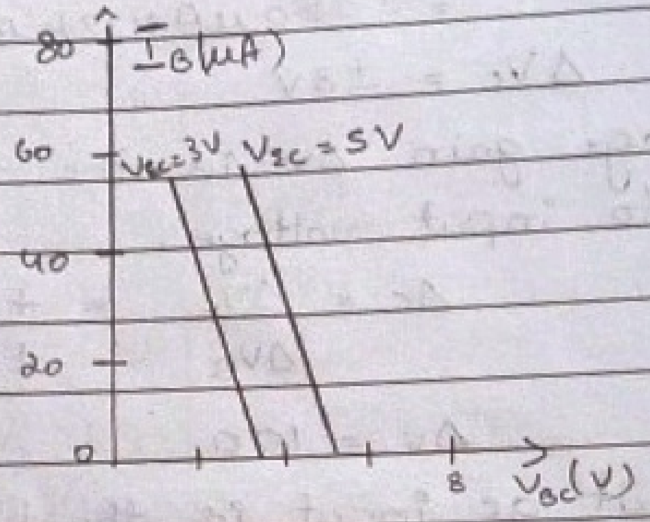
7) Briefly explain common collector input characteristics

Sol Input characteristics are the relationship between the input current and input voltage keeping output voltage constant.

Here input current is IB and input voltage VBE and output voltage is VCE.

The output voltage VCE initially kept at 3V & kept constant. The input voltage VBE is increased from zero gradually & the corresponding input current I_B is noted.

Then the output voltage which is kept constant is increased, let's say 5V & kept constant & output voltage is gradually increased & input current is noted. Graph is drawn with all the values noted.



input characteristics of common collector

9) Explain the transfer and drain characteristics of a JFET.

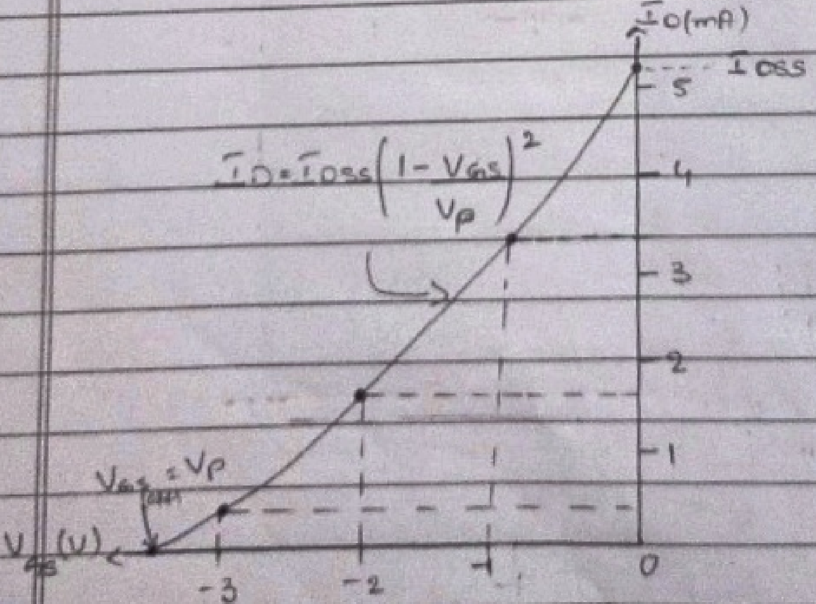
Sol Transfer characteristics of JFET

The transfer characteristics can be determined by observing different values of I_D with variation in V_{GS} provided that the V_{DS} should be constant as shown in figure.

Notice that the bottom end of the transfer characteristic curve is at a point on the V_{GS} axis equal to $V_{GS(off)}$, and the top end of the curve is at a point on the I_D axis equal to I_{DSS}

This curve shows that
i) $I_D = 0$; when $V_{GS} = V_{GS(off)}$
ii) $I_D = I_{DSS}$ when $V_{GS} = 0$
iii) The transfer characteristic curve is expressed approximately as

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

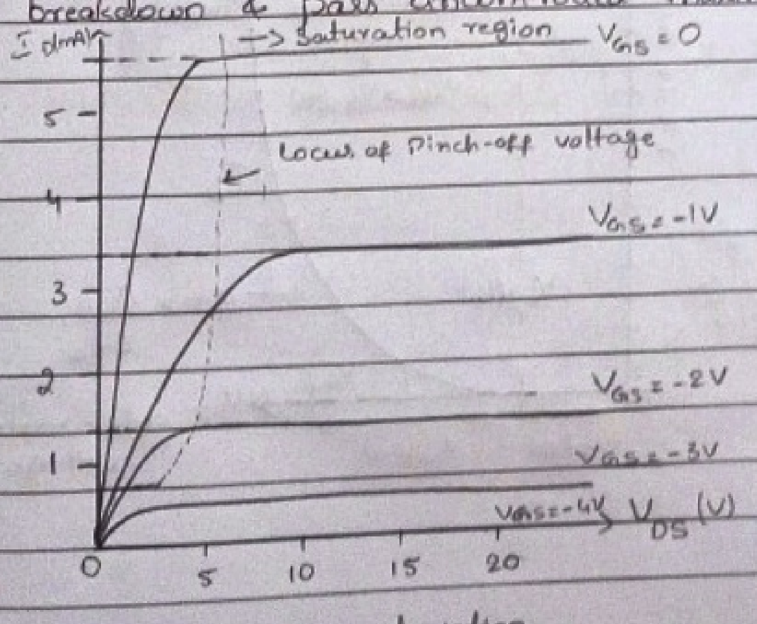


Transfer Characteristics

Drain Characteristics of JFET

The drain characteristics of the JFET were plots between drain current and drain source voltage. These characteristics find the variation of drain current concerning the voltage applied at the drain source terminals while keeping the gate source voltage constant. Basically, the terminologies involved are:-

1. Saturation region - Here, the JFET operates with low Vds & the current is constant, Current is controlled by the gate source voltage.
2. Ohmic region - When $V_{GS}=0$ the depletion region of the channel is very small & JFET acts like a voltage controlled resistor.
3. Cut-off region - This is also known as Pinch-off region where the gate voltage is sufficient to cause the JFET to act as an open circuit as the channel resistance is at maximum.
4. Breakdown region - The voltage b/w the drain & source, is high enough to cause the JFET's resistive channel to breakdown & pass uncontrolled maximum current.



Drain Characteristics

10) Explain the transfer & drain characteristics of enhancement type MOSFET.

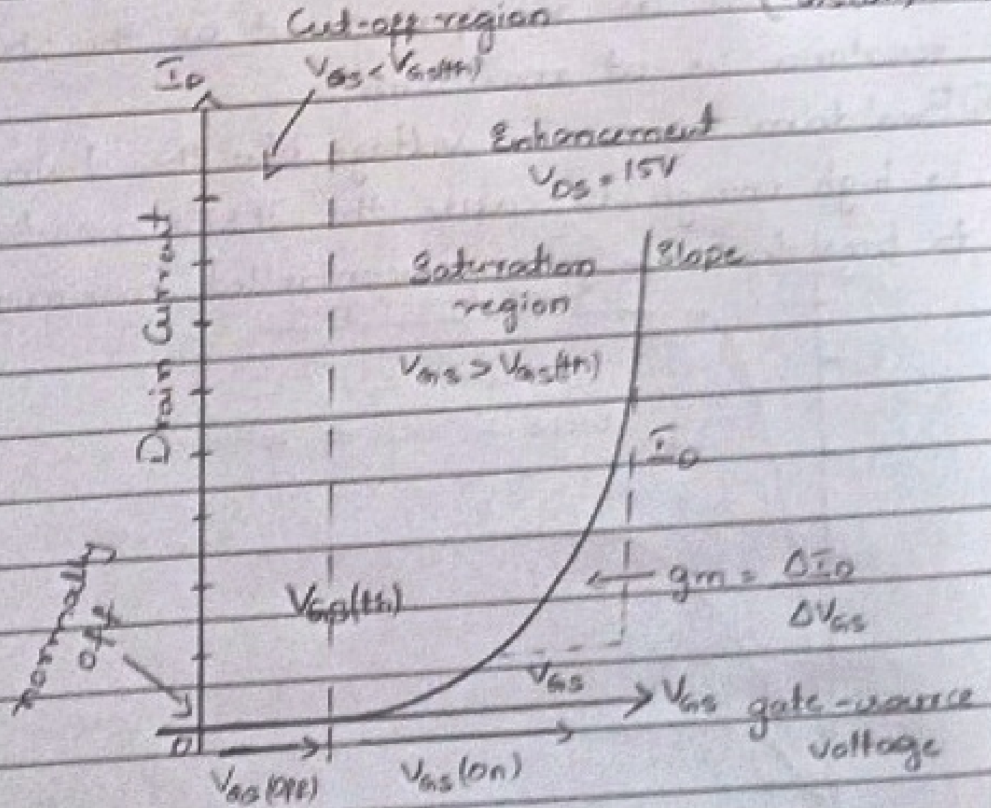
Sol Transfer characteristics of E-MOSFET

i. When $V_{GS} < V_{GS(th)}$, there is no induced channel & the drain current (I_D) is zero.

ii. When $V_{GS} = V_{GS(th)}$, the E-MOSFET is turned on & the induced channel conducts from source to drain.

iii. The further increase in the value of V_{GS} beyond $V_{GS(th)}$ increases the width of induced channel and hence, the drain current.

I_D can be obtained by $I_D = k(V_{GS} - V_{GS(th)})^2$
where $k = \dfrac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$



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Drain Characteristics of E-MOSFET

Drain characteristics is the graph between drain current & drain source voltages for the various positive values of V_{GS}

1) I_D depends on different values of V_{GS}

2) When $V_{GS}=0$, even for large increase in V_{DS}, $I_D=0$
This is said to be cut-off region.